IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: **R26-IT-152**

1. Topic (12 words max)

AI Based - Tourism Safety & Itinerary Platform for Sri Lanka

2. Research group the project belongs to

Centre of Excellence for AI (CEAI)

3. Specialization of the project belongs to

Information Technology – (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sri Lanka's tourism industry has demonstrated strong recovery momentum, recording 2.36 million tourist arrivals in 2025 and setting an ambitious target of 3 million arrivals by 2026. Despite this growth trajectory, the sector continues to face critical structural and technological challenges that threaten sustainability, visitor confidence, and revenue optimization. One of the most pressing issues is the shortage of qualified tourist guides, with an estimated deficit of over 1,600 professionals, leading to inadequate visitor support, poor service quality, and limited destination interpretation.

In parallel, tourist safety has emerged as a major deterrent affecting international trust. Annually, more than 200 tourist fatalities are reported due to road accidents, harassment, and inadequate emergency response mechanisms. These safety concerns disproportionately impact solo female travelers and family tourists, who increasingly prioritize destinations with strong real-time safety assurance systems. The absence of reliable, centralized safety infrastructure significantly weakens Sri Lanka's competitiveness in the global tourism market.

From a technological perspective, existing digital solutions remain fragmented and insufficient. Widely used platforms such as Google Maps lack tourism-specific SOS and predictive safety features, while the official Sri Lanka Tourism mobile application provides only static, informational content without real-time risk assessment, emergency intelligence, or personalized guidance. This disconnect between available data sources and real-time traveler needs results in poor situational awareness and delayed responses during emergencies.

Consequently, Sri Lanka experiences low average daily tourist spending of approximately USD 148, compared to the USD 300 benchmark achieved by regional competitors, contributing to an estimated annual revenue gap of USD 1.6 billion. This underperformance also leads to broader socio-economic consequences, including underemployment within the hospitality sector.

This research addresses these challenges by proposing an AI-driven Natural Language Processing (NLP) platform designed to integrate fragmented tourism, safety, and mobility data into a unified, real-time system. The proposed solution aims to enhance tourist safety, improve decision-making, restore international confidence, and increase overall tourism revenue through intelligent, predictive, and responsive digital intervention.

6. Brief description of Existing Research and Systems (300 words max)

The current landscape of tourism technology in Sri Lanka relies heavily on fragmented, generic platforms.

Popular products like Google Maps provide navigation but lack critical emergency features like an SOS button. TripAdvisor offers community-driven reviews, but these are often static and do not reflect real-time safety conditions.

The official Sri Lanka Tourism app serves primarily as an information portal with static content, failing to provide predictive safety or dynamic assistance. Furthermore, while human guides are essential, there is a documented deficit of 1,600 professionals, forcing reliance on unverified and potentially unsafe alternatives.

From a research perspective, existing systems have explored:

- Chatbots: Utilizing platforms like Dialogflow for basic customer service queries.
- Itinerary Planning: Research papers utilizing BERT and other NLP models for automated travel planning.
- Safety Systems: Apps like 'Citizen' in other countries provide SOS features, but these are not localized for the Sri Lankan context or its specific dialects.

Identified Gaps: Despite these advancements, no integrated platform exists in Sri Lanka that combines SOS functionality, real-time itinerary generation, and flight/hotel bookings into a single interface.

Existing research often ignores the specific needs of the Sri Lankan tourism sector, such as offline support for rural areas, dialect-specific voice recognition, and predictive risk modeling based on local harassment patterns and road safety. Furthermore, current systems do not utilize flight arrival data to provide immediate, safe transit and accommodation bundles for high-risk arrival times.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is an integrated NLP-driven platform that unifies safety features with personalized travel logistics. Unlike traditional travel apps, this system utilizes a unified database consisting of 10,000 hotel records, flight schedules, and tourism site data to address current integration gaps.

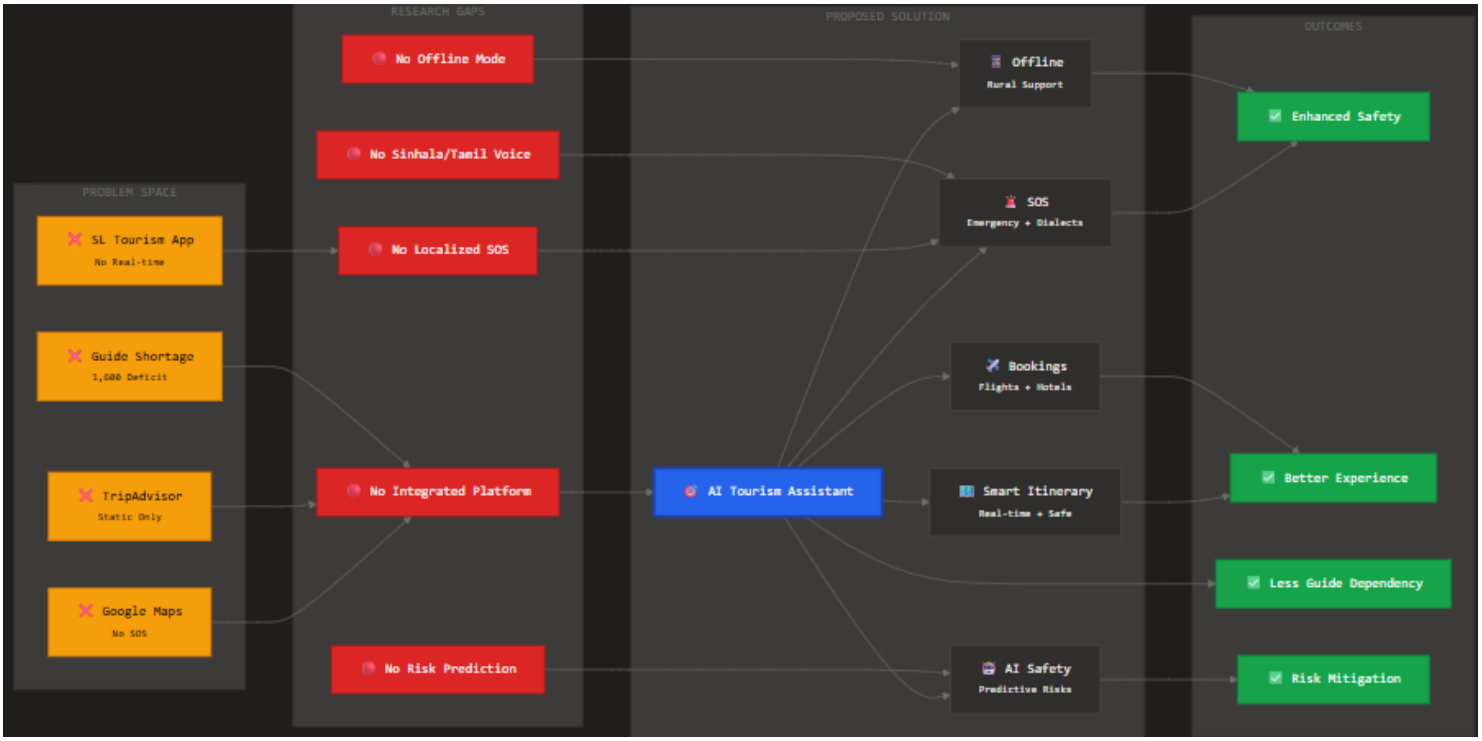
Technical Nature of the System:

Speech & NLP Engine: The system uses Whisper Speech-to-Text and Named Entity Recognition (NER) to process tourist voice queries, including support for Sinhala dialects.

Predictive Safety: By analyzing 100,000 geo-tagged social posts and historical data, the system predicts risks 30 minutes in advance and identifies "danger zones" to avoid.

Smart Itineraries: It generates "Safety-First" itineraries that bundle verified hotels and flights, specifically designed to function offline in rural/jungle areas using AR-guided navigation.

Emergency Intervention: A voice-activated SOS detector connects users directly to police or emergency services with precise GPS coordinates, ensuring protection even without manual phone operation.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The project leverages a robust foundation of domain expertise, technical knowledge, and diverse datasets to ensure the successful development of the AI Tourism Safety & Itinerary Platform.

Domain Expertise:
Our team has access to comprehensive domain insights through several channels:

Governmental & Institutional Reports: We are utilizing official SLTDA Year-In-Review (2024) reports and the Comprehensive Study of Sri Lankan Tour Guides to understand the structural gaps in the tourism industry.

Stakeholder Interviews: Primary domain knowledge is being gathered through structured interviews with local stakeholders, including licensed tour guides and hotel managers in the Negombo and Colombo regions.

Strategic Alignment: The project aligns with global standards by integrating Sustainable Development Goals (SDGs), specifically focusing on SDG 8 (Decent Tourism Jobs), SDG 9 (Digital Infrastructure), and SDG 11 (Safe and Sustainable Tourism).

Technical Knowledge & Infrastructure:
The execution of the solution is supported by high-level technical resources:

NLP & ML Frameworks: Technical implementation is guided by HuggingFace NLP tutorials and specialized research papers on BERT-based itinerary generators.

University Labs: We have access to the Centre of Excellence for AI (CEAI) and specialized AR/ML labs at SLIIT UNI to develop the offline AR navigation and SOS detection systems.

Speech Processing: Expertise in Whisper speech-to-text and NER (Named Entity Recognition) is utilized to handle multilingual queries, including Sinhala dialects.

Data Availability:
The platform relies on a multi-dimensional data collection strategy:

Logistics & Hospitality Data: A dataset of 10,000 hotels (sourced from Booking.com for pricing, locations, and safety ratings) and coordinates for 500+ tourism sites from SLTDA is already accessible.

Real-time Feeds: Live flight schedules and delay statistics are sourced via the AviationStack API to facilitate flight-hotel matching.

Safety & Sentiment Data: To map "danger zones," we are utilizing 100,000 geo-tagged social media posts (e.g., from X/Twitter) to track harassment and safety incidents.

Ethics & Privacy: The project follows strict ethical guidelines, ensuring anonymized tourist data, mandatory SOS consent, and regular bias audits on risk-prediction models.

This combination of verified institutional data, real-time logistics, and high-level AI research infrastructure ensures that the project is both feasible and technically sound for the final-year requirement

9. Objectives and Novelty

Main Objective **Develop AI platform with SOS, safe itineraries using hotel / flight / tourism data to boost Sri Lanka tourism safety / spending.**			
Member Name with Registration No	Sub Objective	Tasks	Novelty
De Silva D.S.K IT22108654	**SOS Safety System**	1. Collect hotel safety reviews + harassment posts. 2. Record flight arrival danger zones. 3. Train "help" voice detection 10 languages. 4. Auto-connect GPS to police/hotels. 5. Test SOS speed in Colombo.	1. Sinhala dialect SOS. 2. Predicts risk 30min ahead. 3. Anonymous police dispatch. 4. Offline jungle SOS. 5. Culture-sensitive responses.
Wanniarachchi P.M.R IT22225092	**Itinerary Generator**	1. Collect 10K hotels + 500 sites locations. 2. Get CMB flight schedules/delays. 3. Generate routes avoiding danger zones. 4. Add hotel bookings to itineraries. 5. Enable AR route guides offline.	1. Safety-first routing with risks. 2. Flight→hotel→site bundles. 3. No-internet AR maps. 4. Budget + safety matching. 5. Learns from user feedback.
De Silva D.C.M IT22108586	**Hotel-Flight Matcher**	1. Collect CMB flight schedules + delay stats. 2. Map hotels within 30min safe drive from airport. 3. Collect verified tourist guides/drivers data (licenses, ratings, SLTDA registered).	1. Flight-delay + verified driver matching. 2. SLTDA-licensed guides/drivers only. 3. Real-time driver ratings from tourists.

		4. Filter 24hr hotels + safe drivers with women/family ratings. 5. Auto-book verified driver pickup with live tracking + SOS.	4. Women/family safe driver priority. 5. Auto-SOS if driver doesn't show.
Abinaya R **IT22090058**	**Tourism Data Analyzer**	1. Collect sites + hotel occupancy data. 2. Track flight tourist volumes. 3. Find overcrowding patterns. 4. Predict safe busy times. 5. Update itineraries dynamically.	1. Overcrowd avoidance. 2. Flight-based crowd forecast. 3. Auto-reroutes safely. 4. Green site focus. 5. Improves from real trips.

8. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Mr	Buddika	Harshanath	11/01/2026
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				